1 Vectors and Matrices

Linear algebra is about vectors $\boldsymbol{v}$ and matrices A. Those are the basic objects that we can add and subtract and multiply (when their shapes match correctly). The first vector $\boldsymbol{v}$ has two components $v_1 = 2$ and $v_2 = 4$. The vector $\boldsymbol{w}$ is also 2-dimensional.

$$\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \qquad \boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \qquad \boldsymbol{v}+\boldsymbol{w} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

The **linear combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$ are the vectors** $c\boldsymbol{v} + d\boldsymbol{w}$ for all numbers c and d:

$$\textbf{The linear combinations} \quad c\begin{bmatrix} 2 \\ 4 \end{bmatrix} + d\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2c+1d \\ 4c+3d \end{bmatrix} \quad \textbf{fill the } \boldsymbol{xy} \textbf{ plane}$$

The **length** of that vector $\boldsymbol{w}$ is $||\boldsymbol{w}|| = \sqrt{10}$, the square root of $w_1^2 + w_2^2 = 1 + 9$. The **dot product** of $\boldsymbol{v}$ and $\boldsymbol{w}$ is $\boldsymbol{v}\cdot\boldsymbol{w} = v_1w_1 + v_2w_2 = (2)(1) + (4)(3) = 14$. In Section 1.2, $\boldsymbol{v}\cdot\boldsymbol{w}$ will reveal the angle between those vectors.

The big step in Section 1.3 is to introduce a **matrix**. This matrix A contains our two column vectors. The vectors have two components, so the matrix is 2 by 2:

$$A = \begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

When a matrix multiplies a vector, we get a combination $c\boldsymbol{v} + d\boldsymbol{w}$ of its columns:

$$A \text{ times} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 2c+1d \\ 4c+3d \end{bmatrix} = c\boldsymbol{v} + d\boldsymbol{w}.$$

And when we look at **all combinations** $A\boldsymbol{x}$ (with every c and d), those vectors produce the **column space of the matrix** A. Here that column space is a plane.

With three vectors, the new matrix B has columns $\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{z}$. In this example $\boldsymbol{z}$ is a combination of $\boldsymbol{v}$ and $\boldsymbol{w}$. So the column space of B is still the xy plane. The vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ are **independent** but $\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{z}$ are **dependent**. A combination produces zero:

$$B = \begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} & \boldsymbol{z} \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 3 & 7 \end{bmatrix} \text{ has } B\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \boldsymbol{v} + \boldsymbol{w} - \boldsymbol{z} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The final goal is to understand **matrix multiplication AB = A times each column of B.**

1.1 Vectors and Linear Combinations

1 $2\boldsymbol{v} - 3\boldsymbol{w}$ **is a linear combination** $c\boldsymbol{v} + d\boldsymbol{w}$ of the vectors $\boldsymbol{v}$ and $\boldsymbol{w}$.

2 For $\boldsymbol{v} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ that combination is $2\begin{bmatrix} 4 \\ 1 \end{bmatrix} - 3\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

3 All combinations $c\begin{bmatrix} 4 \\ 1 \end{bmatrix} + d\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ fill the xy plane. They produce every $\begin{bmatrix} x \\ y \end{bmatrix}$.

4 The vectors $c\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + d\begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$ fill a **plane** in xyz space. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ **is not on that plane.**

Calculus begins with numbers x and functions $f(x)$. Linear algebra begins with vectors $\boldsymbol{v}, \boldsymbol{w}$ and their linear combinations $c\boldsymbol{v} + d\boldsymbol{w}$. Immediately this takes you into two or more (possibly many more) dimensions. But linear combinations of vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ are built from just two basic operations:

Multiply a vector $\boldsymbol{v}$ by a number $\quad 3\boldsymbol{v} = 3\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$

Add vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ of the same dimension: $\boldsymbol{v} + \boldsymbol{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$

Those operations come together in a **linear combination** $c\boldsymbol{v} + d\boldsymbol{w}$ of $\boldsymbol{v}$ and $\boldsymbol{w}$:

Linear combination
$c = 5$ **and** $d = -2$ $\quad 5\boldsymbol{v} - 2\boldsymbol{w} = 5\begin{bmatrix} 2 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix} - \begin{bmatrix} 8 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

That idea of a linear combination opens up two key questions:

1 **Describe all the combinations $c\boldsymbol{v} + d\boldsymbol{w}$.** Do they fill a plane or a line?

2 **Find the numbers c and d** that produce a specific combination $c\boldsymbol{v} + d\boldsymbol{w} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

We can answer those questions here in Section 1.1. But linear algebra is not limited to 2 vectors in 2-dimensional space. As long as we stay linear, the problems can get bigger (more dimensions) and harder (more vectors). The vectors will have m components instead of 2 components. We will have n vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$ instead of 2 vectors. **Those n vectors in m-dimensional space will go into the columns of an m by n matrix A:**

m rows
n columns
m by n matrix

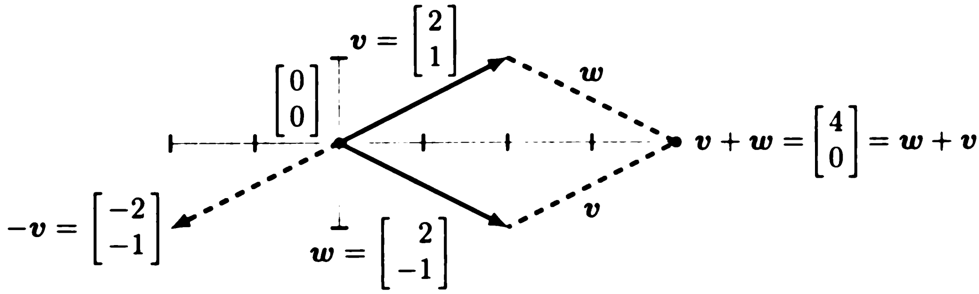
$$A = \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{bmatrix}$$

Let me repeat the two key questions using A, and then retreat back to $m = 2$ and $n = 2$:

1	**Describe all the combinations $A\boldsymbol{x} = x_1\boldsymbol{v}_1 + x_2\boldsymbol{v}_2 + \cdots + x_n\boldsymbol{v}_n$ of the columns**
2	**Find the numbers x_1 to x_n** that produce a desired output vector $A\boldsymbol{x} = \boldsymbol{b}$

Linear Combinations $cv + dw$

Start from the beginning. A vector $\boldsymbol{v}$ in 2-dimensional space has *two components*. To draw $\boldsymbol{v}$ and $-\boldsymbol{v}$, use an arrow that begins at the zero vector:



The vectors $c\boldsymbol{v}$ (for all numbers c) fill an infinitely long line in the xy plane. If $\boldsymbol{w}$ is not on that line, then the vectors $d\boldsymbol{w}$ fill a second line. We aim to see that **the linear combinations $c\boldsymbol{v} + d\boldsymbol{w}$ fill the plane**. Combining points on the $\boldsymbol{v}$ line and the $\boldsymbol{w}$ line gives all points.

Here are four different linear combinations—*we can choose any numbers c and d*:

$$\begin{aligned} 1\boldsymbol{v} + 1\boldsymbol{w} &= \text{sum of vectors} \\ 1\boldsymbol{v} - 1\boldsymbol{w} &= \text{difference of vectors} \\ 0\boldsymbol{v} + 0\boldsymbol{w} &= \textbf{\textit{zero vector}} \\ c\boldsymbol{v} + 0\boldsymbol{w} &= \text{vector } c\boldsymbol{v} \text{ in the direction of } \boldsymbol{v} \end{aligned}$$

Solving Two Equations

Very often linear algebra offers the choice of a picture or a computation. The picture can start with $\boldsymbol{v} + \boldsymbol{w}$ and $\boldsymbol{w} + \boldsymbol{v}$ (those are equal). Another important vector is $\boldsymbol{w} - \boldsymbol{v}$, going backwards on $\boldsymbol{v}$. The Preface has a picture with many more combinations—starting to fill a plane. But if we aim for a particular vector like $c\boldsymbol{v} + d\boldsymbol{w} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$, it will be better to compute the exact numbers c and d. Here are two ways to write down this problem.

$$\boxed{\text{Solve} \quad c\begin{bmatrix} 2 \\ 1 \end{bmatrix} + d\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}. \qquad \text{This means} \quad \begin{aligned} \mathbf{2c + 2d} &= \mathbf{8} \\ \mathbf{c - d} &= \mathbf{2} \end{aligned}.}$$

The rules for solution are simple but strict. We can multiply equations by numbers (*not zero!*) and we can subtract one equation from another equation. Experience teaches that the key is to produce zeros on the left side of the equations. One zero will be enough!

$\frac{1}{2}$(equation 1) is $\boldsymbol{c + d = 4}$

Subtract this from equation 2

Then c is eliminated

$$\begin{aligned} \mathbf{2c + 2d} &= \mathbf{8} \\ \mathbf{0c - 2d} &= \mathbf{-2} \end{aligned}$$

The second equation gives $\boldsymbol{d = 1}$. Going upwards, the first equation becomes $\mathbf{2c + 2 = 8}$. Its solution is $\boldsymbol{c = 3}$. This combination is correct:

$$\mathbf{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \mathbf{1}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}. \quad \text{In matrix form} \begin{bmatrix} \mathbf{2} & \mathbf{2} \\ \mathbf{1} & \mathbf{-1} \end{bmatrix}\begin{bmatrix} \mathbf{3} \\ \mathbf{1} \end{bmatrix} = \begin{bmatrix} \mathbf{8} \\ \mathbf{2} \end{bmatrix}$$

Column Way, Row Way, Matrix Way	
Column way **Linear combination**	$c\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + d\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$
Row way **Two equations for c and d**	$\begin{aligned} v_1c + w_1d &= b_1 \\ v_2c + w_2d &= b_2 \end{aligned}$
Matrix way **2 by 2 matrix**	$\begin{bmatrix} v_1 & w_1 \\ v_2 & w_2 \end{bmatrix}\begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

If the points $\boldsymbol{v}$ and $\boldsymbol{w}$ and the zero vector $\mathbf{0}$ are not on the same line, there is exactly one solution c, d. Then the linear combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$ exactly fill the xy plane. The vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ are **"linearly independent"**. The 2 by 2 matrix $A = \begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix}$ is *"invertible"*.

Can Elimination Fail?

Elimination fails to produce a solution only when the equations don't have a solution in the first place. This can happen when the vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ *lie on the same line through the center point* $(0, 0)$. Those vectors are not independent.

The reason is clear: All combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$ will then *lie on that same line.* If the desired vector $\boldsymbol{b}$ is *off the line*, then the equations $c\boldsymbol{v} + d\boldsymbol{w} = \boldsymbol{b}$ have no solution:

$$\textbf{Example} \qquad \boldsymbol{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \boldsymbol{w} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} \qquad \boldsymbol{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \begin{aligned} \mathbf{1}c + \mathbf{3}d &= \mathbf{1} \\ \mathbf{2}c + \mathbf{6}d &= \mathbf{0} \end{aligned}$$

Those two equations can't be solved. To eliminate the 2 in the second equation, we multiply equation 1 by 2. Then elimination subtracts $2c + 6d = \mathbf{2}$ from the second equation $2c + 6d = \mathbf{0}$. **The result is $\mathbf{0 = -2}$:** *impossible.* We not only eliminated c, we also eliminated $\boldsymbol{d}$.

With $\boldsymbol{v}$ and $\boldsymbol{w}$ on the same line, combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$ fill that line but not a plane. When $\boldsymbol{b}$ is not on that line, no combination of $\boldsymbol{v}$ and $\boldsymbol{w}$ equals $\boldsymbol{b}$. The original vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ are **"linearly dependent"** because $\boldsymbol{w} = 3\boldsymbol{v}$.

Linear combinations of two *independent vectors* $\boldsymbol{v}$ and $\boldsymbol{w}$ in two-dimensional space can produce any vector $\boldsymbol{b}$ in that plane. Then these equations have a solution:

$$\begin{matrix} \text{2 equations} \\ \text{2 unknowns} \end{matrix} \qquad c\boldsymbol{v} + d\boldsymbol{w} = \boldsymbol{b} \qquad \begin{aligned} cv_1 + dw_1 &= b_1 \\ cv_2 + dw_2 &= b_2 \end{aligned}$$

Summary The combinations $c\boldsymbol{v} + d\boldsymbol{w}$ fill the x-y plane unless $\boldsymbol{v}$ is in line with $\boldsymbol{w}$.

Important Here is a different example where elimination seems to be in trouble. But we can easily fix the problem and go on.

$$c\begin{bmatrix} \mathbf{0} \\ \mathbf{1} \end{bmatrix} + d\begin{bmatrix} \mathbf{2} \\ \mathbf{3} \end{bmatrix} = \begin{bmatrix} \mathbf{2} \\ \mathbf{7} \end{bmatrix} \qquad \text{or} \qquad \begin{aligned} 0 + 2d &= 2 \\ c + 3d &= 7 \end{aligned}$$

That zero looks dangerous. But we only have to **exchange equations to find $\boldsymbol{d}$ and $\boldsymbol{c}$:**

$$\begin{aligned} c + 3d &= 7 \\ 0 + 2d &= 2 \end{aligned} \qquad \text{leads to} \quad \boldsymbol{d} = 1 \ \text{ and } \ \boldsymbol{c} = 4$$

Vectors in Three Dimensions

Suppose $\boldsymbol{v}$ and $\boldsymbol{w}$ have three components instead of two. Now they are vectors in three-dimensional space (which we will soon call $\mathbf{R}^3$). We still think of points in the space and arrows out from the zero vector. And we still have linear combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$:

$$\boldsymbol{v} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \quad \boldsymbol{w} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \boldsymbol{v} + \boldsymbol{w} = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} \quad c\boldsymbol{v} + d\boldsymbol{w} = \begin{bmatrix} 2c + d \\ 3c + d \\ c + 0 \end{bmatrix}$$

But there is a difference! The combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$ **do not fill** the whole 3-dimensional space. If we only have 2 vectors, their combinations can at most fill a 2-dimensional plane. It is not a case of linear dependence, it is just a case of not enough vectors.

We need *three* independent vectors if we want their combinations to fill 3-dimensional space. Here is the simplest choice $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ for those three independent vectors:

$$\boldsymbol{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \boldsymbol{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \boldsymbol{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad c\boldsymbol{i} + d\boldsymbol{j} + e\boldsymbol{k} = \begin{bmatrix} c \\ d \\ e \end{bmatrix} \tag{1}$$

Now $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ go along the x, y, z axes in three-dimensional space $\mathbf{R}^3$. We can easily write any vector $\boldsymbol{v}$ in $\mathbf{R}^3$ as a combination of $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$:

Vector form
Matrix form

$$\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k} \qquad \begin{bmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

That is the 3 by 3 **identity matrix** $\boldsymbol{I}$. Multiplying by I leaves every vector unchanged. I is the matrix analog of the number 1, because $I\boldsymbol{v} = \boldsymbol{v}$ for every $\boldsymbol{v}$.

Notice that we could not take a linear combination of a 2-dimensional vector $\boldsymbol{v}$ and a 3-dimensional vector $\boldsymbol{w}$. They are not in the same space.

How Do We Know It is a Plane?

Suppose $\boldsymbol{v}$ and $\boldsymbol{w}$ are nonzero vectors with three components each. Assume they are *independent*, so the vectors point in different directions: $\boldsymbol{w}$ is not a multiple $c\boldsymbol{v}$. *Then their linear combinations fill a plane inside* 3*-dimensional space.* The surface is flat. Here is one way to see that this is true:

> Look at any two combinations $c\boldsymbol{v} + d\boldsymbol{w}$ and $C\boldsymbol{v} + D\boldsymbol{w}$. Halfway between those points is $\boldsymbol{h} = \frac{1}{2}(c + C)\boldsymbol{v} + \frac{1}{2}(d + D)\boldsymbol{w}$. This is another combination of $\boldsymbol{v}$ and $\boldsymbol{w}$. So our surface has the basic property of a plane: Halfway between any two points on the surface is **another point $\boldsymbol{h}$ on the surface**. The surface must be flat!
>
> Maybe that reasoning is not complete, even with the exclamation point. We depended on intuition for the properties of a plane. Another proof is coming in Section 1.2.